Theory of Elementary Particles
Lecture 7

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17.11.2026, Wrocław



Reduction of the product of representations

Isospin symmetry

Quark Model and SU(3)

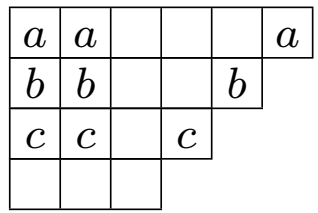
Baryons and Mesons within SU(3)

Reduction of the product of representations

$$\mathbf{8} \times \mathbf{8} = \begin{array}{|c|c|}\hline a & a \\ \hline b \\ \cline{1-1}\end{array} \times \begin{array}{|c|c|}\hline & \\ \hline \\ \cline{1-1}\end{array} = ? \qquad (1)$$

product reduction

- In the tableau for the first factor, assign the same symbol, say a, to all boxes in the first row, the same b to all the boxes in the second row, etc.



- Attach boxes labelled by the symbol a to a tableau of the second factor in all possible way, subject to the rules that no two a's appear in the same column and the resultant graph is still a Young tableau (the length of rows does not increase going from top to bottom and there are not more than n rows, etc.). Repeat this process with bs, cs etc.
- After all symbols have been added to the tableau, these added symbols are then read from right to left in the first row, then the second row,..., and so forth. This sequence of symbols say $aabbac$...must be a lattice permutation. Thus, to the left of any symbol there are no fewer a than b and no fewer b than c etc.

* no lattice: baa; lattice: aaabbc

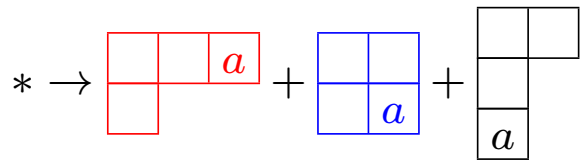
Lattice permutation according to Wiki:

- In mathematics, a lattice word (or lattice permutation) is a string composed of positive integers, in which every prefix contains at least as many positive integers i as integers $i + 1$.
- 11122121 is a lattice permutation 12122111 is not a lattice permutation, since the sub-word 12122 contains more two's than one's.

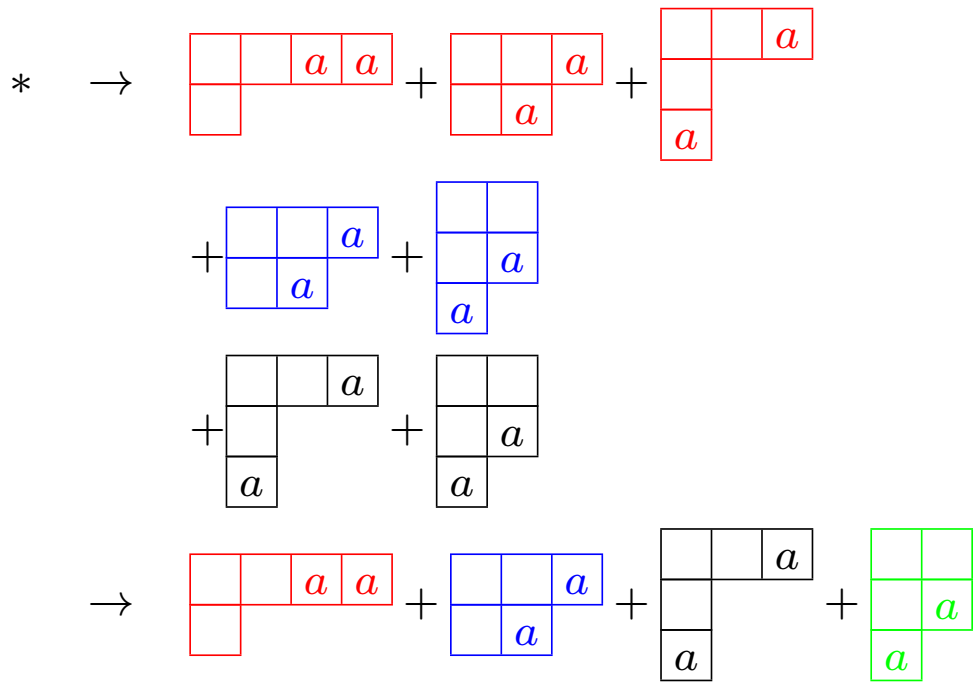
Example $SU(3)$

- $$\mathbf{8} \times \mathbf{8} = \begin{array}{|c|c|}\hline a & a \\ \hline b \\ \cline{1-1}\end{array} \times \begin{array}{|c|c|}\hline & \\ \hline \\ \cline{1-1}\end{array} = ? \qquad (2)$$

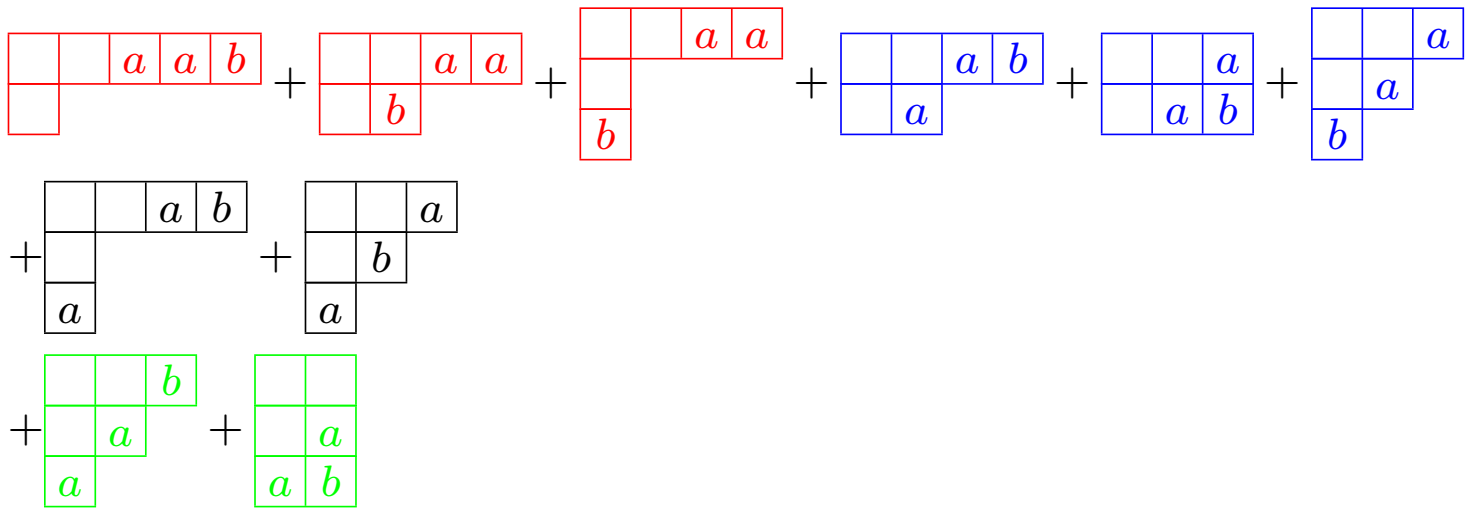
- first a

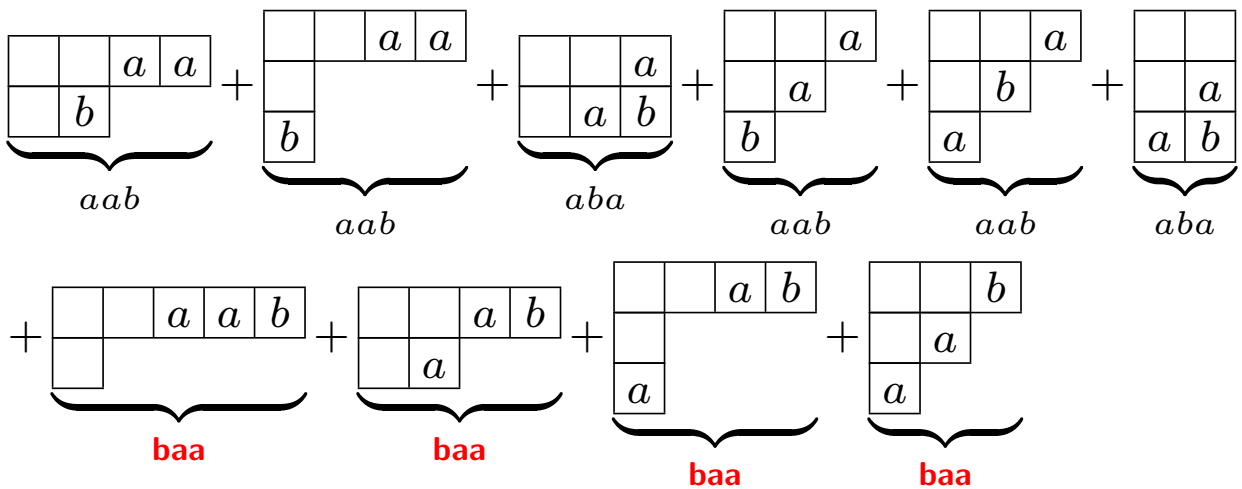


- second a



- b

$*$ $\rightarrow$ 

$*$ $=$ 

$$* \quad = \quad \underbrace{\begin{array}{|c|c|c|c|}\hline \ & \ & a & a \\ \hline \ & b \\ \cline{1-2}\end{array}}_{\mathbf{27}} + \underbrace{\begin{array}{|c|c|c|c|}\hline \ & \ & a & a \\ \hline b \\ \cline{1-1}\end{array}}_{\mathbf{10}} + \underbrace{\begin{array}{|c|c|c|}\hline \ & \ & a \\ \hline \ & a & b \\ \cline{1-3}\end{array}}_{\mathbf{10^*}} + \underbrace{\begin{array}{|c|c|c|}\hline \ & \ & a \\ \hline \ & a \\ \cline{1-2} b \\ \cline{1-1}\end{array}}_{\mathbf{8}} + \underbrace{\begin{array}{|c|c|c|}\hline \ & \ & a \\ \hline \ & b \\ \cline{1-2} a \\ \cline{1-1}\end{array}}_{\mathbf{8}} + \underbrace{\begin{array}{|c|c|}\hline \ & \ \\ \hline \ & a \\ \hline a & b \\ \cline{1-2}\end{array}}_{\mathbf{1}}$$

hence

$$\mathbf{8 \times 8 = 1 + 8 + 8 + 10 + 10^* + 27}$$

Isospin symmetry

- Charge symmetry: apart from electromagnetic and weak interactions, mesons and baryons behave in exactly the same way as their charged symmetric counterparts.

Examples

- Near equality of the proton and neutron masses and the small difference in the nuclear binding energies of ^3H and ^3He (0.8 MeV out of 8 MeV)
- Spectra of light nuclei ^7Li, ^7Be or ^{11}B and ^{11}C show levels with identical angular momenta at approximately the same relative energies $\rightarrow$ Symmetry of the system under interchange neutrons with protons.
- In the isospin-symmetric limit, the strong nucleon–nucleon interaction is invariant under the interchange of protons and neutrons.
- Charge independence implies that the strong nucleon–nucleon interaction is approximately invariant under rotations in isospin space.

- Strong interaction – no difference between proton and neutron – nucleon can be either a proton or neutron... Abstract space spanned by two vectors: neutron and proton states.
- Charge independence can then be viewed as rotational invariance in charge space, and rotations in this space can be generated as in ordinary space by an operator called isobaric spin or isospin
- Isospin invariance is also approximate because it is related to the approximate equality of the u and d quark masses.

- The nucleon state can be described by eight component field

$$\psi(x) = \begin{pmatrix} \psi_p(x) \\ \psi_n(x) \end{pmatrix} \tag{3}$$

- Let us introduce the Baryon number operator

$$\begin{aligned} N_B &= \int d^3x :\psi^\dagger(x)\psi(x): & (4) \\ &= \int d^3x \left(:\psi_p^\dagger(x)\psi_p(x): + :\psi_n^\dagger(x)\psi_n(x):\right) \\ &= (N_p + N_n) - (N_{\overline{p}} + N_{\overline{n}}) & (5) \end{aligned}$$

-

$$N_p = \int \frac{d^3p}{(2\pi)^3} \sum_s a_p^\dagger(\mathbf{p}, s) a_p(\mathbf{p}, s) \quad N_{\overline{p}} = \int \frac{d^3p}{(2\pi)^3} \sum_s b_p^\dagger(\mathbf{p}, s) b_p(\mathbf{p}, s) \tag{6}$$

Notice that

$$\begin{aligned} * &= \int d^3x \frac{d^3p d^3k}{\sqrt{4E_pE_k}(2\pi)^6} \sum_{s,s'} u^\dagger(\mathbf{k}, s') u(\mathbf{p}, s) e^{i(E_k - E_p)t - i\mathbf{x}\cdot(\mathbf{k}-\mathbf{p})} \ldots & (7) \\ &= \int \frac{d^3p}{2E_p(2\pi)^3} \sum_{s,s'} 2E_p \delta_{s,s'} \cdots = \int \frac{d^3p}{(2\pi)^3} \sum_s \cdots & (8) \end{aligned}$$

- Charge operator

$$Q = N_p - N_{\overline{p}} \tag{9}$$

Explanation

$$\star = \int d^3x \psi^\dagger(x)\psi(x) \tag{10}$$

$$= \int d^3x \int \frac{d^3\mathbf{p}}{\sqrt{2E_p}(2\pi)^3} \sum_s \left(a^\dagger(\mathbf{p},s)u^\dagger(\mathbf{p},s)e^{\mathrm{i}p\cdot x} + b(\mathbf{p},s)v^\dagger(\mathbf{p},s)e^{-\mathrm{i}p\cdot x}\right)$$

$$\times \int \frac{d^3\mathbf{q}}{\sqrt{2E_q}(2\pi)^3} \sum_{s'} \left(a(\mathbf{q},s')u(\mathbf{q},s')e^{-\mathrm{i}q\cdot x} + b^\dagger(\mathbf{q},s')v(\mathbf{q},s')e^{\mathrm{i}q\cdot x}\right) \tag{11}$$

$$= \sum_{s,s'} \int \frac{d^3\mathbf{p}}{2E_p(2\pi)^3} \left\{a^\dagger(\mathbf{p},s)a(\mathbf{p},s')u^\dagger(\mathbf{p},s)u(\mathbf{p},s') + b(\mathbf{p},s)b^\dagger(\mathbf{p},s')v^\dagger(\mathbf{p},s)v(\mathbf{p},s')\right\}$$

$$+ \sum_{s,s'} \int \frac{d^3\mathbf{p}}{2E_p(2\pi)^3} a^\dagger(\mathbf{p},s)b^\dagger(-\mathbf{p},s')u^\dagger(\mathbf{p},s)v(-\mathbf{p},s')e^{i2E_pt}$$

$$+ \sum_{s,s'} \int \frac{d^3\mathbf{p}}{2E_p(2\pi)^3} b(\mathbf{p},s)a(-\mathbf{p},s')v^\dagger(\mathbf{p},s)u(-\mathbf{p},s')e^{-i2E_pt} \tag{12}$$

Notice that

$$\sum_{s,s'} u^\dagger(\mathbf{p},s)v(-\mathbf{p},s') = \sum_{s,s'} v^\dagger(\mathbf{p},s)u(-\mathbf{p},s') = 0 \tag{13}$$

$$\star \quad = \quad \int \sum_s \frac{d^3\mathbf{p}}{(2\pi)^3} \left\{ a^\dagger(\mathbf{p},s) a(\mathbf{p},s) + b(\mathbf{p},s) b^\dagger(\mathbf{p},s) \right\} \tag{14}$$

$$N_p \mid p\rangle = \mid p\rangle, \quad N_n \mid p\rangle = 0 \tag{15}$$
$$N_n \mid n\rangle = \mid n\rangle, \quad N_p \mid n\rangle = 0 \tag{16}$$

So N_B and Q can not be considered to be an isospin operator for the nucleon field but the linear combination

$$T_3 = Q - \frac{1}{2}N_B = \frac{1}{2}(N_p - N_{\overline{p}}) - \frac{1}{2}(N_n - N_{\overline{n}}) \tag{17}$$

then

$$T_3 = \frac{1}{2}\int d^3x\, \psi^\dagger \tau_3 \psi \tag{18}$$

It suggests the identification with the third Pauli matrix in the isospin space.

- Let us introduce the isospin operator

$$T_i = \frac{1}{2}\int d^3x\, \psi^\dagger \tau_i \psi, \quad i = 1,2,3 \tag{19}$$

- Isospin invariance means that the strong interaction Hamiltonian H_h is such that it commutes with each T_i:

$$[\mathbf{T}, H_h] = 0 \tag{20}$$

  i.e. $\mathbf{T}$ is conserved vector.

- $\mathbf{T}$ generates the rotations of the field ψ in isospin space. Indeed

$$[T_i, \psi_A(x)] = -\frac{1}{2}\tau_{i\,AB}\psi_B(x) \quad [T_i, \psi^\dagger_A(x)] = +\frac{1}{2}\tau_{i\,BA}\psi^\dagger_B(x) \tag{21}$$

  (exercise) use relation

$$\{\psi^\dagger_a(x_0, \mathbf{x}), \psi_b(x_0, \mathbf{y})\} = \delta_{ab}\delta^{(3)}(\mathbf{x} - \mathbf{y}) \tag{22}$$

- Consider real unit vector $\boldsymbol{\theta} = (\theta_1, \theta_2, \theta_3)$
-

$$\psi + i[\boldsymbol{\theta}\cdot\mathbf{T}, \psi] = \psi - \frac{i}{2}\boldsymbol{\theta}\cdot\boldsymbol{\tau}\psi \tag{23}$$

$$\psi + i\boldsymbol{\theta}\cdot\mathbf{T}\psi - i\psi\boldsymbol{\theta}\cdot\mathbf{T} = \left(1 - \frac{i}{2}\boldsymbol{\theta}\cdot\boldsymbol{\tau}\right)\psi \tag{24}$$

$$(1 + i\boldsymbol{\theta}\cdot\mathbf{T})\psi(1 - i\boldsymbol{\theta}\cdot\mathbf{T}) = \left(1 - \frac{i}{2}\boldsymbol{\theta}\cdot\boldsymbol{\tau}\right)\psi \tag{25}$$

- The result may then be cast into

$$\mathcal{U}\psi(x)\mathcal{U}^{-1} = S(\boldsymbol{\theta})\psi(x) \tag{26}$$

where

$$\mathcal{U} = \exp(i\boldsymbol{\theta}\cdot\mathbf{T}), \quad S(\boldsymbol{\theta}) = \exp\left(-\frac{i}{2}\boldsymbol{\theta}\cdot\boldsymbol{\tau}\right) \tag{27}$$

- Special case

$$\mathcal{U}_2 = e^{i\pi\tau_2/2}, \quad \mathcal{U}_2\begin{pmatrix}\psi_p\\ \psi_n\end{pmatrix}\mathcal{U}_2^{-1} = \begin{pmatrix}-\psi_n\\ \psi_p\end{pmatrix}$$

- Eventually

$$[T_k, T_m] = i\epsilon_{kmn}T_n \tag{28}$$

(exercise)

Isospin SU(2)

Strong interactions are invariant under a transformation which interchanges proton with neutron. Indeed proton and neutron form $SU(2)$ isodublet.

$$T_3 \mid p\rangle = \frac{1}{2} \mid p\rangle, \quad T_3 \mid n\rangle = -\frac{1}{2} \mid n\rangle \tag{29}$$

and

$$T_- \mid p\rangle = \mid n\rangle, \quad T_+ \mid n\rangle = \mid p\rangle \tag{30}$$

and

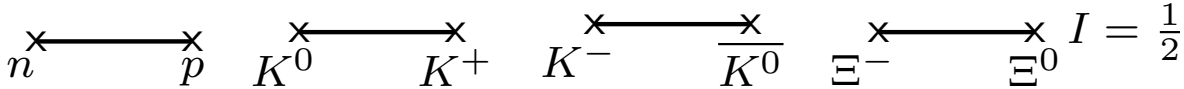
$$[H_{strong}, T_i] = 0, \quad \mathcal{U} H_{strong} \mathcal{U}^{-1} = H_{strong} \tag{31}$$

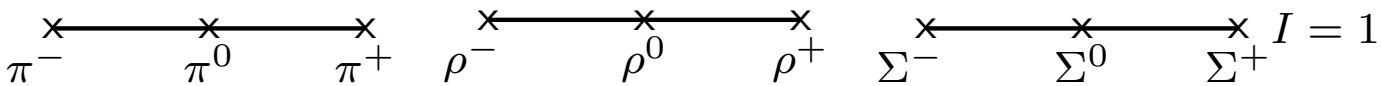
hence

$$\langle p \mid H_{strong} \mid p\rangle = E_p = \langle p \mid \mathcal{U}_2^{-1} \mathcal{U}_2 H_{strong} \mathcal{U}_2^{-1} \mathcal{U}_2 \mid p\rangle = \langle n \mid H_{strong} \mid n\rangle \tag{32}$$

$M_p = M_n$ in the exact isospin limit.

The isospin can be extended to other hadrons:





etc.

- Electromagnetic interactions do not respect the isospin symmetry!
- The difference between quark masses u and d is also a source of the isospin violation.
- Experimentally the mass difference in every multiplet is a good measure of isospin symmetry breaking:

$$\frac{m_n - m_p}{m_n + m_p} \approx 0.7 \times 10^{-3}, \tag{33}$$

$$\frac{m_{\pi^+} - m_{\pi^0}}{m_{\pi^+} + m_{\pi^0}} \approx 1.7 \times 10^{-2} \tag{34}$$

Pion system: $\mathbf{2} \otimes \mathbf{2}^*$ in tensor method:

$$\pi^a_b = \psi^a \psi_b = \frac{1}{2}\delta^a_b \psi^c \psi_c + \left(\psi^a \psi_b - \frac{1}{2}\delta^a_b \psi^c \psi_c\right) \tag{35}$$

- Red tensor: dim=**1** it is Kronecker delta, so it is irreducible representation

$$\frac{1}{2}(\psi^1 \psi_1 + \psi^2 \psi_2) \tag{36}$$

- Blue tensor: dim=**3**

$$\pi^1_2 = \psi^1 \psi_2 \equiv \phi'_1, \tag{37}$$
$$\pi^2_1 = \psi^2 \psi_1 \equiv \phi'_2, \tag{38}$$
$$\pi^1_1 = \frac{1}{2}(\psi^1 \psi_1 - \psi^2 \psi_2) = -\frac{1}{2}(\psi^2 \psi_2 - \psi^1 \psi_1) = -\pi^2_2 \equiv \phi'_3 \tag{39}$$

Hence:

$$\pi^a_b = \begin{pmatrix} \phi'_3 & \phi'_1 \\ \phi'_2 & -\phi'_3 \end{pmatrix} \tag{40}$$

$$\pi^a_b = \begin{pmatrix} \phi'_3 & \phi'_1 \\ \phi'_2 & -\phi'_3 \end{pmatrix} \tag{41}$$

Notice that

$$\psi_1 = \psi^{*1}, \quad \psi_2 = \psi^{*2} \tag{42}$$

Hence

$$\phi'_1 = \phi'^*_2 \equiv (\phi_1 - i\phi_2)/\sqrt{2} \tag{43}$$

and

$$\phi'_3 = \phi'^*_3 = \phi_3/\sqrt{2} \tag{44}$$

As the result

$$\pi^a_b = \frac{1}{\sqrt{2}} \begin{pmatrix} \phi_3 & \phi_1 - i\phi_2 \\ \phi_1 + i\phi_2 & -\phi_3 \end{pmatrix} = \tau_k \phi_k/\sqrt{2} \tag{45}$$

From exercise (list 4) we know that

$$I_3\pi^1_2 = (+1)\pi^1_2, \quad I_3\pi^1_1 = (0)\pi^1_1, \quad I_3\pi^2_1 = (-1)\pi^2_1 \tag{46}$$

π^a_b is three dimensional adjoint representation of $SU(2)$

Let us keep the normalization so that

$$\pi^a_b \pi^b_a = \frac{1}{2} \sum \phi_i \phi_j \mathrm{Tr}(\tau_i \tau_j) = \sum \phi_i^2 \tag{47}$$

then we have the following assignments

- $I_3 = +1: \quad \pi^1{}_2 = \pi^+,$
- $I_3 = \ \ 0: \quad \pi^1{}_1 = -\pi^2{}_2 = \pi^0/\sqrt{2},$
- $I_3 = -1: \quad \pi^2{}_1 = \pi^-.$

and

$$\pi^a_b = \begin{pmatrix} \frac{1}{\sqrt{2}}\pi_0 & \pi^+ \\ \pi^- & -\frac{1}{\sqrt{2}}\pi_0 \end{pmatrix} \tag{48}$$

Quark Model and SU(3): strangeness

- Strangeness S is conserved in strong interactions
- $S(\Lambda^0) = -1$, $S(K^0) = +1$
- S is like an electric charge and would be associated with $U(1)$ symmetry
- Gell-Mann-Nishijima relation

$$Q = T_3 + \frac{Y}{2} \tag{49}$$

with

$$Y = B + S \tag{50}$$

Y is the hypercharge, B is the Baryon number
  - proton: $Q = 1$, $B = 1$, $S = 0$
- there exists a higher symmetry group which contains $SU_{isospin}(2) \times U_Y(1)$ as a subgroup: $SU(3)$

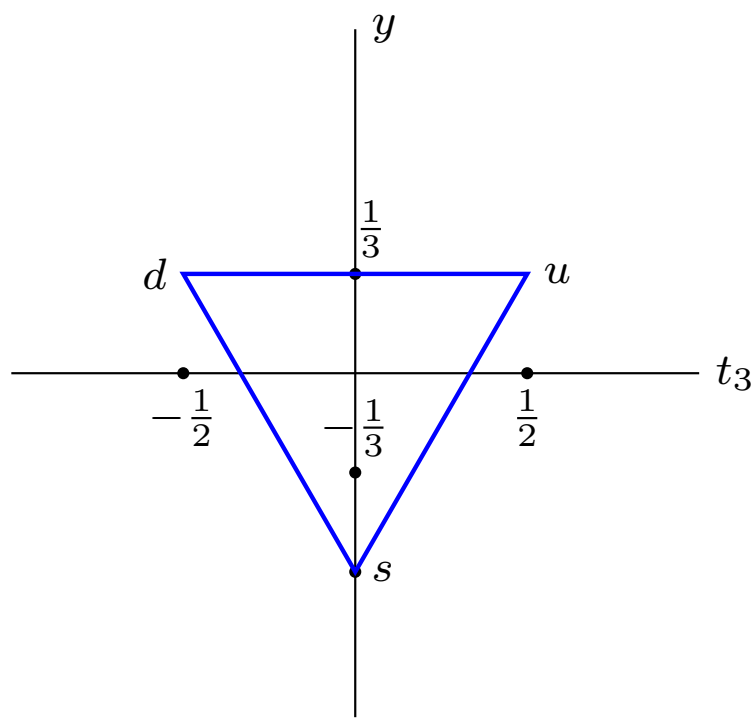
- Hadrons are made of quarks
- in $SU(3)$ flavour group, quarks form basis of the fundamental $SU(3)$ group

$$q_i = \begin{pmatrix} u \\ d \\ s \end{pmatrix}$$

- Baryon consists of three quarks, hence

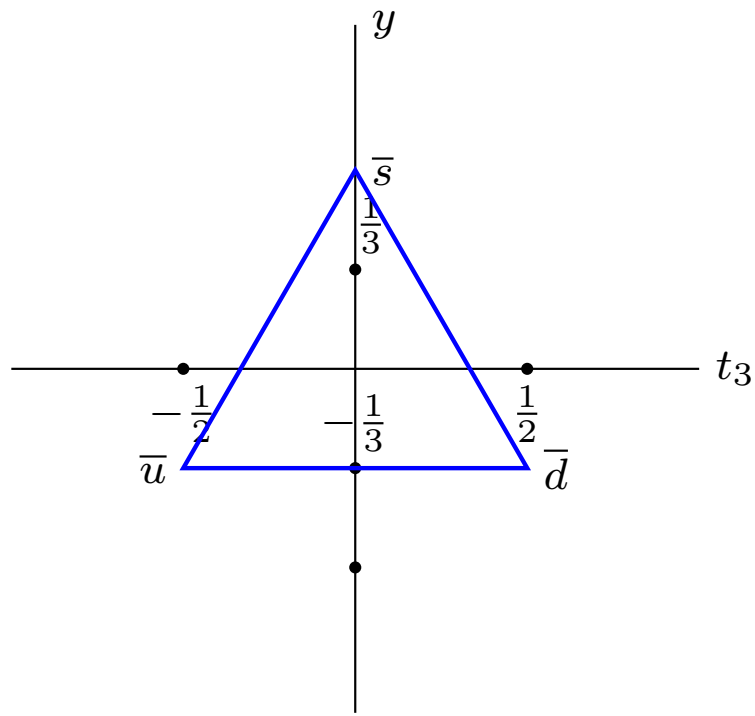
$$B_q = \frac{1}{3}$$

quark	Q	S	Y	T_3
u	$\frac{2}{3}$	0	$\frac{1}{3}$	$\frac{1}{2}$
d	$-\frac{1}{3}$	0	$\frac{1}{3}$	$-\frac{1}{2}$
s	$-\frac{1}{3}$	-1	$-\frac{2}{3}$	0



$(p, q) = (1, 0) = \mathbf{3}$ (triplet)

quark	Q	S	Y	T_3
$\overline{u}$	$-\frac{2}{3}$	0	$-\frac{1}{3}$	$-\frac{1}{2}$
$\overline{d}$	$\frac{1}{3}$	0	$-\frac{1}{3}$	$\frac{1}{2}$
$\overline{s}$	$\frac{1}{3}$	1	$\frac{2}{3}$	0



$(p, q) = (0, 1) = \mathbf{3}^*$ (triplet)

- Mesons

$$q\overline{q} = \square \otimes \begin{array}{|c|}\hline\ \\\hline\ \\\hline\end{array} = \begin{array}{|c|}\hline\ \\\hline\ \\\hline\ \\\hline\end{array} \oplus \begin{array}{|c|c|}\hline\ & \\\hline\ & \\\hline\end{array} = \mathbf{1} + \mathbf{8} \tag{51}$$

- Baryons

$$\begin{aligned} qqq &= \square \otimes \square \otimes \square \\ &= \left(\begin{array}{|c|c|}\hline\ & \\\hline\end{array} \oplus \begin{array}{|c|}\hline\ \\\hline\ \\\hline\end{array} \right) \otimes \square \\ &= \begin{array}{|c|c|c|}\hline\ & & \\\hline\end{array} \oplus \begin{array}{|c|c|}\hline\ & \\\hline\ & \\\hline\end{array} \oplus \begin{array}{|c|c|}\hline\ & \\\hline\ & \\\hline\end{array} \oplus \begin{array}{|c|}\hline\ \\\hline\ \\\hline\ \\\hline\end{array} \\ &= \mathbf{10} + \mathbf{8} + \mathbf{8} + \mathbf{1} \end{aligned}$$

Baryons and Mesons within SU(3)

- Finding the irreducible representations (tensors) If reducible

$$\delta^{b_j}_{a_i} T^{a_1 \ldots a_n}_{b_1 \ldots b_m} \qquad (n,m) \to (n-1, m-1) \tag{52}$$

$$\epsilon_{a_i a_j b_{m+1}} T^{a_1 \ldots a_n}_{b_1 \ldots b_m} \qquad (n,m) \to (n-2, m+1) \tag{53}$$

$$\epsilon^{b_i b_j a_{n+1}} T^{a_1 \ldots a_n}_{b_1 \ldots b_m} \qquad (n,m) \to (n+1, m-2) \tag{54}$$

Consider baryon:

$$\mathbf{3} \otimes \mathbf{3} \otimes \mathbf{3} = (\mathbf{6} \oplus \mathbf{3}^*) \otimes \mathbf{3} = \mathbf{1} \oplus \mathbf{8} \oplus \mathbf{8} \oplus \mathbf{10} \tag{55}$$

Our goal is to construct explicit three-quark flavor wave functions and identify the two octets corresponding to different permutation symmetries.
The corresponding tensor product reads

$$q^i q^j q^n, \tag{56}$$

where q^i is the fundamental representation of $SU(3)$, $i, j, n = 1, 2, 3$.

Convention

In this lecture the quark (fundamental) representation carries an **upper** index, $q^i \sim \mathbf{3}$, and antiquarks carry lower indices — the opposite placement to the tensor-method section of Lecture 6.

- Second rank tensor

$$q^i q^j \tag{57}$$

can be decomposed into symmetric and antisymmetric.

- Indeed it belongs to $\mathbf{3} \otimes \mathbf{3}$ and after symmetrization and anti-symmetrization:

$$\begin{aligned} q^i q^j &= \frac{1}{2}\left(q^i q^j + q^j q^i\right) + \frac{1}{2}\left(\delta^i_n \delta^j_m - \delta^i_m \delta^j_n\right) q^n q^m & (58)\\ &= \frac{1}{2}\left(q^i q^j + q^j q^i\right) + \frac{1}{2}\epsilon^{ijk}\epsilon_{knm} q^n q^m, & (59) \end{aligned}$$

where $\epsilon_{123} = 1$.

- Hence we have obtained 6-dimensional symmetric tensor

$$S^{ij} = \frac{1}{2}\left(q^i q^j + q^j q^i\right) \tag{60}$$

and 3-dimensional antisymmetric

$$\theta_i = \epsilon_{ijk} q^j q^k \tag{61}$$

-

$$q^i q^j = S^{ij} + \frac{1}{2}\epsilon^{ijk}\theta_k \tag{62}$$

- The tensor product $3 \otimes 3 \otimes 3$ can first be decomposed into two reducible tensor products:

$$S^{ij} q^n \quad \text{and} \quad \theta_k q^n \tag{63}$$

$$6 \otimes 3 = 10 \oplus 8, \qquad \overline{3} \otimes 3 = 1 \oplus 8.$$

- The first tensor above has rank 3.

$$S^{ij} q^n \epsilon_{ijn} = 0, \quad S^{ij} q^n \epsilon_{ijk} = 0, \quad \underline{S^{ij} q^n \epsilon_{kjn} \neq 0.} \tag{64}$$

A non-vanishing contraction with the Levi-Civita tensor indicates that $S^{ij} q^n$ contains a mixed-symmetry component.

Reduction of the tensor $S^{ij}q^n$

- Is the tensor $S^{ij}q^n$ irreducible?
- First, notice that
$$\epsilon_{ijn}S^{ij}q^n = 0, \tag{65}$$
since $S^{ij} = S^{ji}$.
- Nevertheless, the tensor is reducible:
$$\mathbf{6} \otimes \mathbf{3} = \mathbf{10} \oplus \mathbf{8}. \tag{66}$$
Its totally symmetric component belongs to the decuplet, while the remaining mixed-symmetry component transforms as an octet.
- To identify the mixed-symmetry component, introduce the tensor
$$T^{mli} = \epsilon^{mlk}\epsilon_{kjn}S^{ij}q^n = S^{im}q^l - S^{il}q^m. \tag{67}$$
- Notice that $T^{mli} = -T^{lmi}$; hence the totally symmetric component does not contribute.

- On the other hand we have

$$
\begin{aligned}
S^{ij}q^n &= \frac{1}{3}\left(S^{ij}q^n + S^{nj}q^i + S^{in}q^j\right) + \frac{1}{3}\left(S^{ij}q^n - S^{nj}q^i\right) + \frac{1}{3}\left(S^{ij}q^n - S^{in}q^j\right) \\
&= \frac{1}{3}\left(S^{ij}q^n + S^{nj}q^i + S^{in}q^j\right) + \frac{1}{3}S^{lj}q^m\left(\delta^i_l\delta^n_m - \delta^n_l\delta^i_m\right) \\
&\quad + \frac{1}{3}S^{il}q^m\left(\delta^j_l\delta^n_m - \delta^n_l\delta^j_m\right) \qquad (68) \\
&= \frac{1}{3}\left(S^{ij}q^n + S^{nj}q^i + S^{in}q^j\right) + \frac{1}{3}S^{lj}q^m\epsilon_{lmk}\epsilon^{kin} + \frac{1}{3}S^{il}q^m\epsilon_{lmk}\epsilon^{kjn} \qquad (69) \\
&= \frac{1}{3}\left(S^{ij}q^n + S^{nj}q^i + S^{in}q^j\right) + \frac{1}{3}\epsilon_{lmk}\left(S^{lj}q^m\epsilon^{kin} + S^{il}q^m\epsilon^{kjn}\right) \qquad (70)
\end{aligned}
$$

$$
\left(\delta^j_l\delta^n_m - \delta^n_l\delta^j_m\right) = \epsilon^{kjn}\epsilon_{klm} \qquad (71)
$$

- The first tensor is totally symmetric irreducible tensor (contracting with ϵ gives zero).
- In previous slides we have shown that the second tensor is also irreducible.
- It is the tensor symmetric under a change of first two indexes.

$$
\frac{1}{3}\epsilon_{lmk}\left(S^{lj}q^m\epsilon^{kin} + S^{il}q^m\epsilon^{kjn}\right) = (\leftrightarrow) = \frac{1}{3}\epsilon_{lmk}S^{li}q^m\epsilon^{kjn} + \frac{1}{3}\epsilon_{lmk}S^{jl}q^m\epsilon^{kin} \qquad (72)
$$

- Now, consider the tensor

$$\theta_k q^n \tag{73}$$

- Certainly irreducible tensor is given by the trace, namely

$$\theta_n q^n \tag{74}$$

- It is easy to show that the tensor

$$\theta_k q^n - \frac{1}{3}\delta^n_k \theta_i q^i \tag{75}$$

is also irreducible, because its trace vanishes

$$\theta_i q^i - \frac{1}{3}\cdot 3 \cdot \theta_i q^i = 0 \tag{76}$$

- Now we are ready to decompose $\theta_k q^n$ tensor as follows

$$\theta_k q^n = \frac{1}{3}\delta^n_k \theta_i q^i + \left(\theta_k q^n - \frac{1}{3}\delta^n_k \theta_i q^i\right) \tag{77}$$

and eventually

$$\epsilon^{ijk}\theta_k q^n = \frac{1}{3}\epsilon^{ijn}\theta_i q^i + \epsilon^{ijk}\left(\theta_k q^n - \frac{1}{3}\delta^n_k \theta_i q^i\right) \tag{78}$$

The first irreducible tensor is totally antisymmetric, while the second is antisymmetric under a change of first two indexes.

Eventually, the third rank tensor decomposes as follows

$$q^i q^j q^n = \underbrace{\frac{1}{3}\left(S^{ij}q^n + S^{nj}q^i + S^{in}q^j\right)}_{\mathbf{10}} + \underbrace{\frac{1}{3}\epsilon_{lmk}\left(S^{lj}q^m\epsilon^{kin} + S^{il}q^m\epsilon^{kjn}\right)}_{\mathbf{8}}$$

$$+ \underbrace{\frac{1}{6}\epsilon^{ijn}\theta_l q^l}_{\mathbf{1}} + \underbrace{\frac{1}{2}\epsilon^{ijk}\left(\theta_k q^n - \frac{1}{3}\delta^n_k\theta_l q^l\right)}_{\mathbf{8}} \tag{79}$$

$$= \frac{1}{\sqrt{6}}D^{ijn} + \frac{1}{\sqrt{2}}\epsilon^{ijk}B^n_k + \frac{1}{\sqrt{6}}\epsilon^{ijn}S + \frac{2}{\sqrt{6}}\left(\epsilon^{kin}N^j_k + \epsilon^{kjn}N^i_k\right) \tag{80}$$

How many independent components has the tensor

$$\frac{1}{6}\epsilon_{lmk}\left(S^{lj}q^m\epsilon^{kin}+S^{il}q^m\epsilon^{kjn}\right)=\frac{1}{\sqrt{6}}\left(N^j_k\epsilon^{kin}+N^i_k\epsilon^{kjn}\right)? \tag{81}$$

where

$$N^j_k=\frac{1}{\sqrt{6}}\epsilon_{lmk}S^{lj}q^m \tag{82}$$

Then

$$N^j_k = \frac{1}{\sqrt{6}} \begin{pmatrix} \epsilon_{lm1} S^{l1} q^m & \epsilon_{lm2} S^{l1} q^m & \epsilon_{lm3} S^{l1} q^m \\ 2q^2 q^2 q^3 - (q^3 q^2 + q^2 q^3) q^2 & (q^3 q^2 + q^2 q^3) q^1 - (q^1 q^2 + q^2 q^1) q^3 & (q^1 q^2 + q^2 q^1) q^2 - 2q^2 q^2 q^1 \\ (q^2 q^3 + q^3 q^2) q^3 - 2q^3 q^3 q^2 & 2q^3 q^3 q^1 - (q^1 q^3 + q^3 q^1) q^3 & (q^1 q^3 + q^3 q^1) q^2 - (q^2 q^3 + q^3 q^2) q^1 \end{pmatrix} \tag{83}$$

It is convenient to take u as the first quark (i=1), d for the second ($i = 2$) and s for the third.

$$N^1_1 = \epsilon_{lm1} S^{l1} q^m = \epsilon_{231} S^{21} q^3 + \epsilon_{321} S^{31} q^2 \tag{84}$$

$$= S^{21} q^3 - S^{31} q^2 = \frac{1}{2} \left(q^2 q^1 q^3 + q^1 q^2 q^3 - q^3 q^1 q^2 - q^1 q^3 q^2 \right) \tag{85}$$

$$= \frac{1}{2} \left(dus + uds - sud - usd \right) \tag{86}$$

$$N^j_k = \frac{1}{\sqrt{6}} \begin{pmatrix} (du+ud)s-(su+us)d & (us+su)u-2uus & 2uud-(du+ud)u \\ 2dds-(sd+ds)d & (sd+ds)u-(ud+du)s & (ud+du)d-2ddu \\ (ds+sd)s-2ssd & 2ssu-(us+su)s & (us+su)d-(ds+sd)u \end{pmatrix} \quad (87)$$

In $SU(3)$ there exist three doublets: u, d (T); d, s (U) and u, s (V), where

$$T_+ d = u, \quad T_- u = d, \quad T_\pm s = 0, \tag{88}$$

$$U_+ s = d, \quad U_- d = s, \quad U_\pm u = 0, \tag{89}$$

$$V_+ s = u, \quad V_- u = s, \quad V_\pm d = 0, \tag{90}$$

$$\tag{91}$$

In order to extract physical states we take the state of neutron, act on it with the raising isospin operator and obtain the proton, namely we know that in the $1/2$ $SU(2)$ representation

$$T_+ \left| -\frac{1}{2} \right\rangle = \left| \frac{1}{2} \right\rangle \tag{92}$$

$$p \equiv T_+ n = T_+ \frac{1}{\sqrt{6}} \left[(ud + du)d - 2ddu \right] \tag{93}$$

$$= \frac{1}{\sqrt{6}} \left[uud + udu + uud + duu - 2udu - 2duu \right] \tag{94}$$

$$= \frac{1}{\sqrt{6}} \left[2uud - udu - duu \right] \tag{95}$$

And others

$$\Sigma^+ \equiv U_- p = U_- \frac{1}{\sqrt{6}} [2uud - udu - duu] = \frac{1}{\sqrt{6}} [2uus - usu - suu] \quad (96)$$

$$\Xi^0 \equiv V_- \Sigma^+ = V_- \frac{1}{\sqrt{6}} [2uus - usu - suu] = \frac{1}{\sqrt{6}} [sus + uss - 2ssu] \quad (97)$$

$$\Xi^- \equiv T_- \Xi^0 = T_- \frac{1}{\sqrt{6}} [sus + uss - 2ssu] = \frac{1}{\sqrt{6}} [sds + dss - 2ssd] \quad (98)$$

$$\Sigma^- \equiv U_+ \Xi^- = U_+ \frac{1}{\sqrt{6}} [(sds + dss) - 2ssd] \quad (99)$$

$$= \frac{1}{\sqrt{6}} [(dds + sdd + dds + dsd) - 2dsd - 2sdd] \quad (100)$$

$$= \frac{1}{\sqrt{6}} [2dds - dsd - sdd] \quad (101)$$

Then

$$N^j_k = \begin{pmatrix} \frac{1}{\sqrt{6}}\left[(du+ud)s-(su+us)d\right] & -\Sigma^+ & p \\ \Sigma^- & \frac{1}{\sqrt{6}}\left[(sd+ds)u-(ud+du)s\right] & n \\ \Xi^- & -\Xi^0 & \frac{1}{\sqrt{6}}\left[(us+su)d-(ds+sd)u\right] \end{pmatrix} \tag{102}$$

Now take the Σ^+ and reconstruct Σ^0:

$$\begin{aligned} \sqrt{2}\Sigma^0 &= T_-\Sigma^+ && (103)\\ &= T_-\frac{1}{\sqrt{6}}\left[2uus-usu-suu\right] && (104)\\ &= \frac{1}{\sqrt{6}}\left[2dus+2uds-dsu-usd-sdu-sud\right] && (105)\\ &= \frac{1}{\sqrt{6}}\left[2(du+ud)s-(ds+sd)u-(us+su)d\right] && (106) \end{aligned}$$

Now, take the N_1^1 element, which is a sum of Σ^0 and Λ^0, namely

$$\alpha\Sigma^0 + \beta\Lambda^0 = \frac{1}{\sqrt{6}}\left[(du+ud)s - (su+us)d\right] \tag{107}$$

Notice that

$$\alpha = \langle \Sigma^0 \mid \alpha\Sigma^0 + \beta\Lambda^0 \rangle = \frac{1}{\sqrt{2}} \tag{108}$$

Therefore

$$\beta\Lambda^0 = \frac{1}{\sqrt{6}}\left[(du+ud)s - (su+us)d\right] - \frac{1}{\sqrt{2}}\Sigma^0 = \frac{1}{\sqrt{6}}\left[-\frac{1}{2}(su+us)d + \frac{1}{2}(ds+sd)u\right] \tag{109}$$

next

$$\langle \beta\Lambda^0 \mid \beta\Lambda^0 \rangle = \frac{1}{6} \Rightarrow \beta = \frac{1}{\sqrt{6}}. \tag{110}$$

so we get

$$\Lambda^0 = \frac{1}{2}(ds+sd)u - \frac{1}{2}(su+us)d. \tag{111}$$

then

$$N_k^j = \begin{pmatrix} \frac{1}{\sqrt{2}}\Sigma^0 + \frac{1}{\sqrt{6}}\Lambda^0 & -\Sigma^+ & p \\ \Sigma^- & -\frac{1}{\sqrt{2}}\Sigma^0 + \frac{1}{\sqrt{6}}\Lambda^0 & n \\ \Xi^- & -\Xi^0 & -\frac{2}{\sqrt{6}}\Lambda^0 \end{pmatrix} \tag{112}$$

Indeed N_k^j is traceless.

Now consider the tensor

$$B_k^n = \frac{1}{\sqrt{2}}\left(\theta_k q^n - \frac{1}{3}\delta_k^n \theta_i q^i\right) \tag{113}$$

First find the components of θ_i:

$$\begin{aligned}
\theta_1 &= \epsilon_{1jk} q^j q^k = q^2 q^3 - q^3 q^2 = ds - sd && (114)\\
\theta_2 &= \epsilon_{2jk} q^j q^k = q^3 q^1 - q^1 q^3 = su - us && (115)\\
\theta_3 &= \epsilon_{3jk} q^j q^k = q^1 q^2 - q^2 q^1 = ud - du && (116)\\
\frac{1}{3}\theta_i q^i &= \frac{1}{3}\left((ds - sd)u + (su - us)d + (ud - du)s\right) && (117)
\end{aligned}$$

Hence one can compute

$$B^n_k = \frac{1}{\sqrt{2}} \begin{pmatrix} \theta_1 q^1 - \frac{1}{3}\theta_i q^i & \theta_2 q^1 & \theta_3 q^1 \\ \theta_1 q^2 & \theta_2 q^2 - \frac{1}{3}\theta_i q^i & \theta_3 q^2 \\ \theta_1 q^3 & \theta_2 q^3 & \theta_3 q^3 - \frac{1}{3}\theta_i q^i \end{pmatrix} \tag{118}$$

$$= \frac{1}{\sqrt{2}} \begin{pmatrix} (ds - sd)u - \frac{1}{3}\theta_i q^i & (su - us)u & (ud - du)u \\ (ds - sd)d & (su - us)d - \frac{1}{3}\theta_i q^i & (ud - du)d \\ (ds - sd)s & (su - us)s & (ud - du)s - \frac{1}{3}\theta_i q^i \end{pmatrix} \tag{119}$$

$$= \begin{pmatrix} \frac{2(ds-sd)u-(su-us)d-(ud-du)s}{3\sqrt{2}} & \frac{(su-us)u}{\sqrt{2}} & \frac{(ud-du)u}{\sqrt{2}} \\ \frac{(ds-sd)d}{\sqrt{2}} & \frac{2(su-us)d-(ds-sd)u-(ud-du)s}{3\sqrt{2}} & \frac{(ud-du)d}{\sqrt{2}} \\ \frac{(ds-sd)s}{\sqrt{2}} & \frac{(su-us)s}{\sqrt{2}} & \frac{2(ud-du)s-(ds-sd)u-(su-us)d}{3\sqrt{2}} \end{pmatrix} \tag{120}$$

Let the neutron wave function be

$$n \equiv \frac{(ud-du)d}{\sqrt{2}} \tag{121}$$

$$p \equiv T_+ n = \frac{(ud-du)u}{\sqrt{2}} \tag{122}$$

$$\Sigma^+ \equiv U_- p = \frac{(us-su)u}{\sqrt{2}} \tag{123}$$

$$\Xi^0 \equiv V_- \Sigma^+ = \frac{(us-su)s}{\sqrt{2}} \tag{124}$$

$$\Xi^- \equiv T_- \Xi^0 = \frac{(ds-sd)s}{\sqrt{2}} \tag{125}$$

$$\Sigma^- \equiv U_+ \Xi^- = \frac{(ds-sd)d}{\sqrt{2}} \tag{126}$$

$$\Sigma^0 \equiv \frac{1}{\sqrt{2}} T_- \Sigma^+ = \frac{(us-su)d+(ds-sd)u}{2} \tag{127}$$

then

$$B_k^n = \begin{pmatrix} \frac{2(ds-sd)u-(su-us)d-(ud-du)s}{3\sqrt{2}} & -\Sigma^+ & p \\ \Sigma^- & \frac{2(su-us)d-(ds-sd)u-(ud-du)s}{3\sqrt{2}} & n \\ \Xi^- & -\Xi^0 & \frac{2(ud-du)s-(ds-sd)u-(su-us)d}{3\sqrt{2}} \end{pmatrix} \tag{128}$$

On the other hand

$$\frac{2(ds-sd)u-(su-us)d-(ud-du)s}{3\sqrt{2}}=\alpha\Sigma^0+\beta\Lambda^0 \tag{129}$$

hence

$$\alpha=\frac{1}{\sqrt{2}},\quad \beta\Lambda^0=B_1^1-\alpha\Sigma^0=\frac{1}{6\sqrt{2}}\left[(ds-sd)u-2(ud-du)s-(us-su)d\right]. \tag{130}$$

then

$$|\beta|^2=\frac{1}{6}\Rightarrow\beta=\frac{1}{\sqrt{6}} \tag{131}$$

and

$$\Lambda^0=\frac{1}{\sqrt{12}}\left[(ds-sd)u-2(ud-du)s-(us-su)d\right] \tag{132}$$

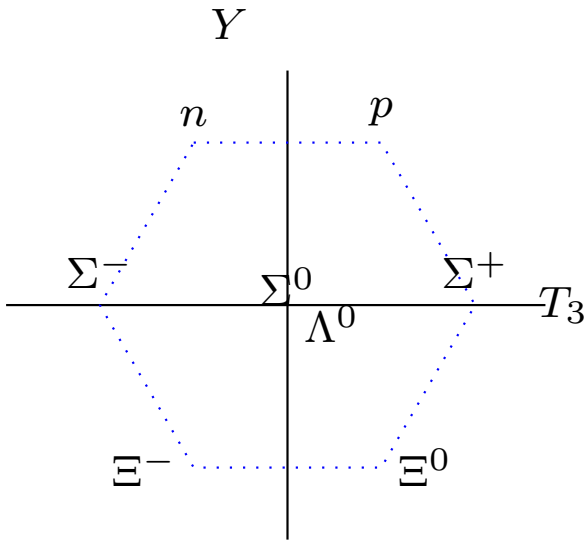
Then

$$B_k^n = \begin{pmatrix} \frac{1}{\sqrt{2}}\Sigma^0 + \frac{1}{\sqrt{6}}\Lambda^0 & -\Sigma^+ & p \\ \Sigma^- & -\frac{1}{\sqrt{2}}\Sigma^0 + \frac{1}{\sqrt{6}}\Lambda^0 & n \\ \Xi^- & -\Xi^0 & -\frac{2}{\sqrt{6}}\Lambda^0 \end{pmatrix} \tag{133}$$

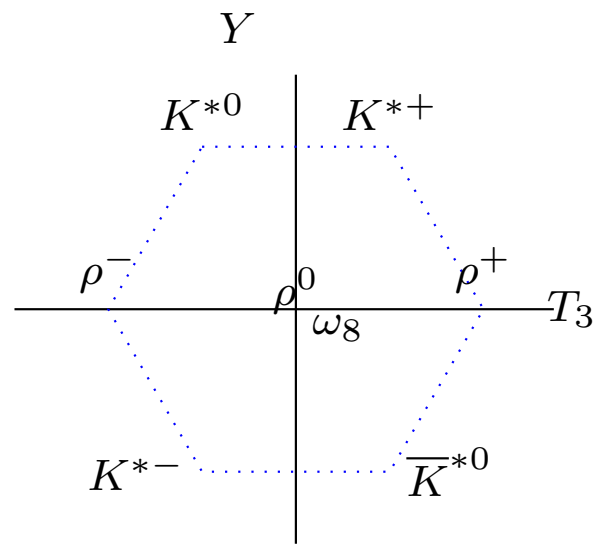
Notice that we have obtained the wave functions of the baryon octet. The mixed symmetric and mixed antisymmetric wave functions.

octet member	$\phi_{M,S}$	$\phi_{M,A}$
p	$\frac{1}{\sqrt{6}}\,[2uud - udu - duu]$	$\frac{1}{\sqrt{2}}(ud - du)u$
n	$\frac{1}{\sqrt{6}}\,[(ud + du)d - 2ddu]$	$\frac{1}{\sqrt{2}}(ud - du)d$
Σ^+	$\frac{1}{\sqrt{6}}\,[2uus - usu - suu]$	$\frac{1}{\sqrt{2}}(us - su)u$
Σ^0	$\frac{1}{\sqrt{12}}\,[2(du + ud)s - (ds + sd)u - (us + su)d]$	$\frac{1}{2}\,[(us - su)d + (ds - sd)u]$
Σ^-	$\frac{1}{\sqrt{6}}\,[2dds - dsd - sdd]$	$\frac{1}{\sqrt{2}}(ds - sd)d$
Ξ^0	$\frac{1}{\sqrt{6}}\,[sus + uss - 2ssu]$	$\frac{1}{\sqrt{2}}(us - su)s$
Ξ^-	$\frac{1}{\sqrt{6}}\,[sds + dss - 2ssd]$	$\frac{1}{\sqrt{2}}(ds - sd)s$
Λ^0	$\frac{1}{2}\,[(ds + sd)u - (su + us)d]$	$\frac{1}{\sqrt{12}}\,[(ds - sd)u - 2(ud - du)s - (us - su)d]$

Table: Baryon octet

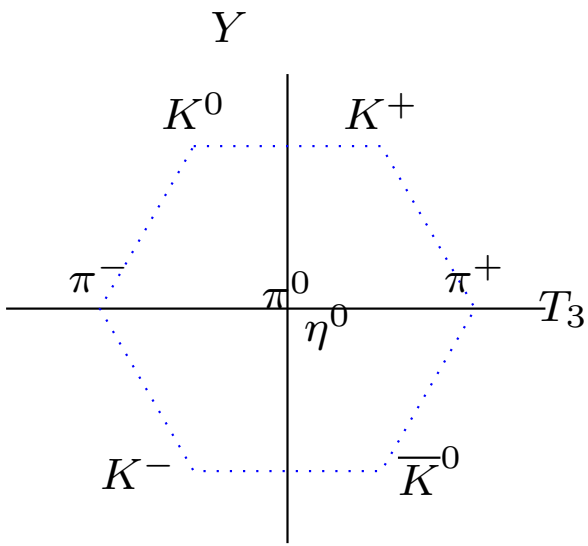


$(1,1) = \mathbf{8}$ (octet), $\frac{1}{2}^+$ baryons

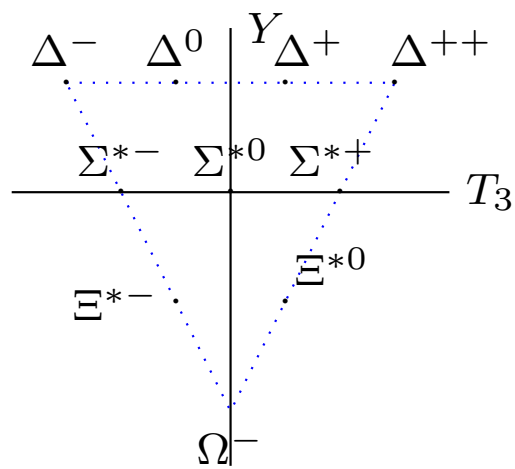


$(1,1) = \mathbf{8}$ (octet), 1^- mesons

The physical ω and ϕ mesons arise from mixing of the SU(3) singlet and octet states.



$(1,1) = \mathbf{8}$ (octet), 0^- mesons



$(3,0) = \mathbf{10}$(decuplet) $\frac{3}{2}^+$ baryons